

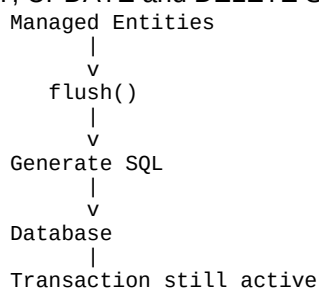
Chapter 10 - flush(), clear(), evict(), refresh() & detach()

1. Introduction

Hibernate provides methods to control the Persistence Context and synchronize managed entities with the database. Understanding these methods is important for debugging and interviews.

2. flush()

flush() synchronizes the current Persistence Context with the database. It executes pending INSERT, UPDATE and DELETE SQL statements but does not commit the transaction.



3. clear()

clear() removes all managed entities from the Persistence Context. All entities become detached. A later access requires fetching entities again.

4. evict()

evict(entity) removes only the specified entity from the Session cache while leaving other managed entities unchanged.

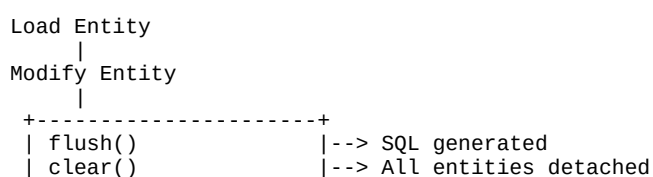
5. refresh()

refresh(entity) reloads the entity from the database and discards any in-memory changes that have not yet been synchronized.

6. detach()

detach(entity) is the JPA API that removes a single entity from the Persistence Context. It is similar in purpose to Hibernate's evict(entity).

7. Execution Flow



```

| evict(entity)          |--> One entity detached
| refresh(entity)        |--> Reload from DB
| detach(entity)         |--> One entity detached (JPA)
+-----+

```

Method	Effect
flush()	Synchronizes pending changes with the database
clear()	Removes all managed entities
evict(entity)	Removes one managed entity
refresh(entity)	Reloads state from the database
detach(entity)	Detaches one entity (JPA)

8. Example

```

Session session=sessionFactory.openSession();
Transaction tx=session.beginTransaction();

Employee emp=session.get(Employee.class,1);
emp.setName("Siva");

session.flush();      // UPDATE SQL executes

session.refresh(emp); // Reload entity

session.evict(emp);   // emp detached

tx.commit();
session.close();

```

9. Best Practices

- flush() does not commit the transaction.
- Use clear() for batch processing to reduce memory usage.
- Use refresh() when another process may have changed the database.
- Use evict()/detach() to stop tracking specific entities.

10. Interview Questions

1. Difference between flush() and commit()? 2. Difference between clear() and evict()? 3. Difference between evict() and detach()? 4. When do you use refresh()? 5. Does flush() commit the transaction?